Paul DeFrain

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Professional Summary

Cybersecurity professional and M.P.S. candidate in Cybersecurity Analytics and Operations at Penn State University with hands-on experience in security monitoring, threat detection, incident investigation, network analysis, vulnerability assessment and Python-based security automation. Developer of SOC-Forge, an independent security operations platform focused on detection engineering, event correlation, attack reconstruction, and analyst investigation workflows. Experienced with Active Directory, digital forensics, SIEM concepts, and translating technical security findings into actionable recommendations.

Cybersecurity Projects

SOC-Forge | Security Operations & Investigation Platform
Python | Git | Windows Event Logs | MITRE ATT&CK

- Developed a Python-based security operations platform for threat detection, event correlation, case management, and incident investigation
- Implemented detection and correlation capabilities for brute-force attacks, password spraying, account lockouts, privilege escalation, RDP activity, scheduled tasks, and other Windows security events
- Built analyst investigation workflows incorporating alerts, timelines, indicators, evidence, hypotheses, attack reconstruction, MITRE ATT&CK coverage, and investigation reporting
- Developed automated testing and validation across security analysis workflows to maintain reliable detection and investigation behavior

Shipco Data Leak Investigation | Penn State Cyber Range
Digital Forensics | Incident Response | Linux | Log Analysis

- Conducted a simulated enterprise data-leak investigation by analyzing centralized system logs, SSH activity, Bash history, web application events, and mail-server evidence to reconstruct attacker activity
- Assessed security weaknesses involving centralized logging, privileged access monitoring, authentication, file auditing, and data loss prevention controls
- Traced compromised credentials, unauthorized file access, privilege escalation, lateral activity, email spoofing, and exfiltration of sensitive data through a typo-squatted external domain
- Developed remediation recommendations including MFA, privileged access management, SIEM alerting, centralized log collection, least privilege, file auditing and DLP controls

Technical Skills

- **Security Operations & Monitoring:** SIEM Log Analysis, Incident Response, Security Event Analysis, Threat Hunting, Anomaly Detection, Detection Engineering, MITRE ATT&CK
- **Networking & Analysis:** Wireshark, Nmap, Scapy, Network Traffic Analysis, Packet Analysis
- **Systems & Identity:** Windows, Linux, Active Directory, Group Policy, RBAC, Least Privilege
- **Forensics:** Volatility, Rekall, Memory Forensics, Indicators of Compromise
- **Programming & Tools:** Python, Bash, Git, GitHub, VS Code

Education

Master of Professional Studies (M.P.S.), Cybersecurity Analytics & Operations
Penn State – World Campus
GPA: 4.0 | Expected December 2026

Graduate Certificate – Information Systems Cybersecurity
Penn State – World Campus
GPA: 4.0 | Awarded August 2025

Bachelor of Science in Business Administration, Management
Kutztown University of Pennsylvania
Awarded December 2020

Professional Experience

Bartender – Lehigh Country Club, Allentown, PA **June 2024 – Present**

- Delivered exceptional client service and operational management in high-trust executive environments while maintaining strict confidentiality, discretion, and professionalism
- Managed end-to-end logistics and inventory control for large-scale private events under pressure

Retail Associate - Buss Paints, Emmaus, PA ***Nov 2021-June 2024***

- Supported commercial and institutional clients (Schools, hospitals) through inventory management, vendor coordination, and fulfillment of time-sensitive operational requirements
- Collaborated with ownership on small business operations, inventory controls, process improvement, and customer requirements

Intern - Milestone Financial Associates, LLC – Macungie, PA **Feb 2020-Apr 2020**
Conducted research and compiled analytical reports on public companies while organizing sensitive financial data, strengthening skills in risk assessment and confidential information management